



Figure 3: Docker desktop images

Docker

Docker is a lightweight DevOps tool that helps to speed up and ease different workflows in your software development life cycle with a combined methodology. A Docker container image is a standalone package that contains all the dependencies required for running an application. It is an open source lightweight tool to create, deploy, and manage containers on a different host operating system using resource isolation features, such as cgroups and Linux kernels.

A container package can have application code, libraries, and configurations. The container engine is installed on the host OS.

Initial release	2013																		
Stable release	20.10.11																		
Written in	Go																		
Licence	Apache License 2.0 Docker Community Edition																		
Website	https://hub.docker.com/ https://www.docker.com/																		
GitHub repository	https://github.com/docker																		
Features	• Easy management of applications • Docker hub – public registry • Uniform packaging • Integrates well with Kubernetes																		
How is it useful in implementing DevOps practices?	OS-level virtualisation, Containers																		
Can we integrate it with Pipeline as Code?	Yes																		
Is a commercial flavour available?	Docker Enterprise Edition																		
Docker commands	<table> <tr> <th>Description</th><th>Command</th></tr> <tr> <td>Log in to the registry</td><td><code>docker login</code></td></tr> <tr> <td>Build an image from the Docker file</td><td><code>docker build -t sampleapp :1.0</code></td></tr> <tr> <td>List all the images</td><td><code>docker images</code></td></tr> <tr> <td>Delete an image</td><td><code>docker rmi imagename</code></td></tr> <tr> <td>Create a container</td><td><code>docker run --name containerName dockerImage</code></td></tr> <tr> <td>Pull an image from the registry</td><td><code>docker pull imageName</code></td></tr> <tr> <td>Push an image</td><td><code>docker push imagename: tag</code></td></tr> <tr> <td>Clean an unused image</td><td><code>docker image prune</code></td></tr> </table>	Description	Command	Log in to the registry	<code>docker login</code>	Build an image from the Docker file	<code>docker build -t sampleapp :1.0</code>	List all the images	<code>docker images</code>	Delete an image	<code>docker rmi imagename</code>	Create a container	<code>docker run --name containerName dockerImage</code>	Pull an image from the registry	<code>docker pull imageName</code>	Push an image	<code>docker push imagename: tag</code>	Clean an unused image	<code>docker image prune</code>
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Figure 4: Google Kubernetes engine

Kubernetes

Kubernetes or k8s is an open source, extensible and portable container orchestration platform developed by Google. It offers declarative configuration and automation for deployment. It has become one of the most popular tools for DevOps practices because of its capabilities and features for deployment, management, and scaling applications in Kubernetes clusters. It helps to manage resources effectively.

We can create a Kubernetes cluster in an on-premise environment as well as on private and public clouds. Public cloud provides managed services for Kubernetes. Google Cloud Platform has Google Kubernetes Engine, AWS has Amazon Elastic Kubernetes Services (EKS) and Azure has Azure Kubernetes Service (AKS) for managing Kubernetes clusters in a cloud environment.

Initial release	2014																
Stable release	1.22.2																
Written in	Go																
Licence	Apache License 2.0																
Website	https://kubernetes.io/																
GitHub repository	https://github.com/kubernetes/Kubernetes																
Features	• Distributed architecture • Easy integration with Docker • Supports hybrid, on-premise, or cloud environments																
How is it useful in implementing DevOps practices?	Container orchestration and management																
Can we integrate it with Pipeline as Code?	Yes																
Is a commercial flavour available?	Cloud flavours are available																
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